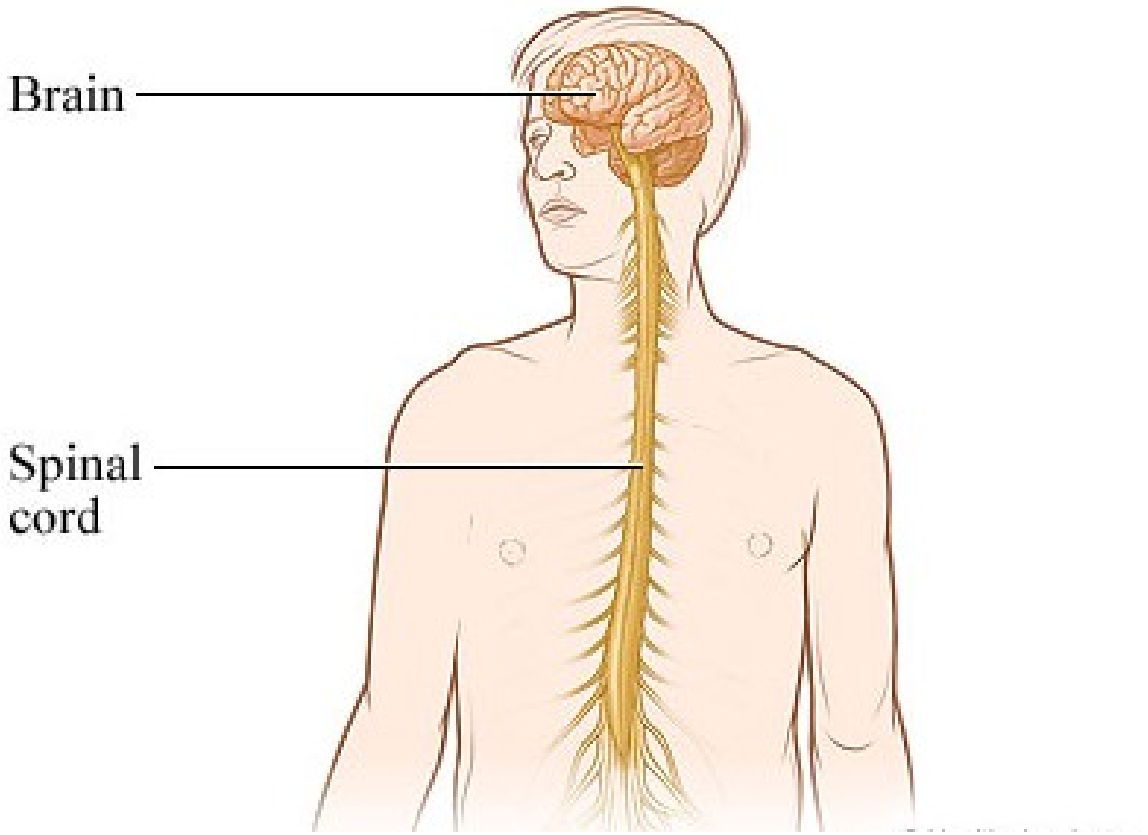
The Central Nervous System (CNS) - protection



The Nervous System is highly organised

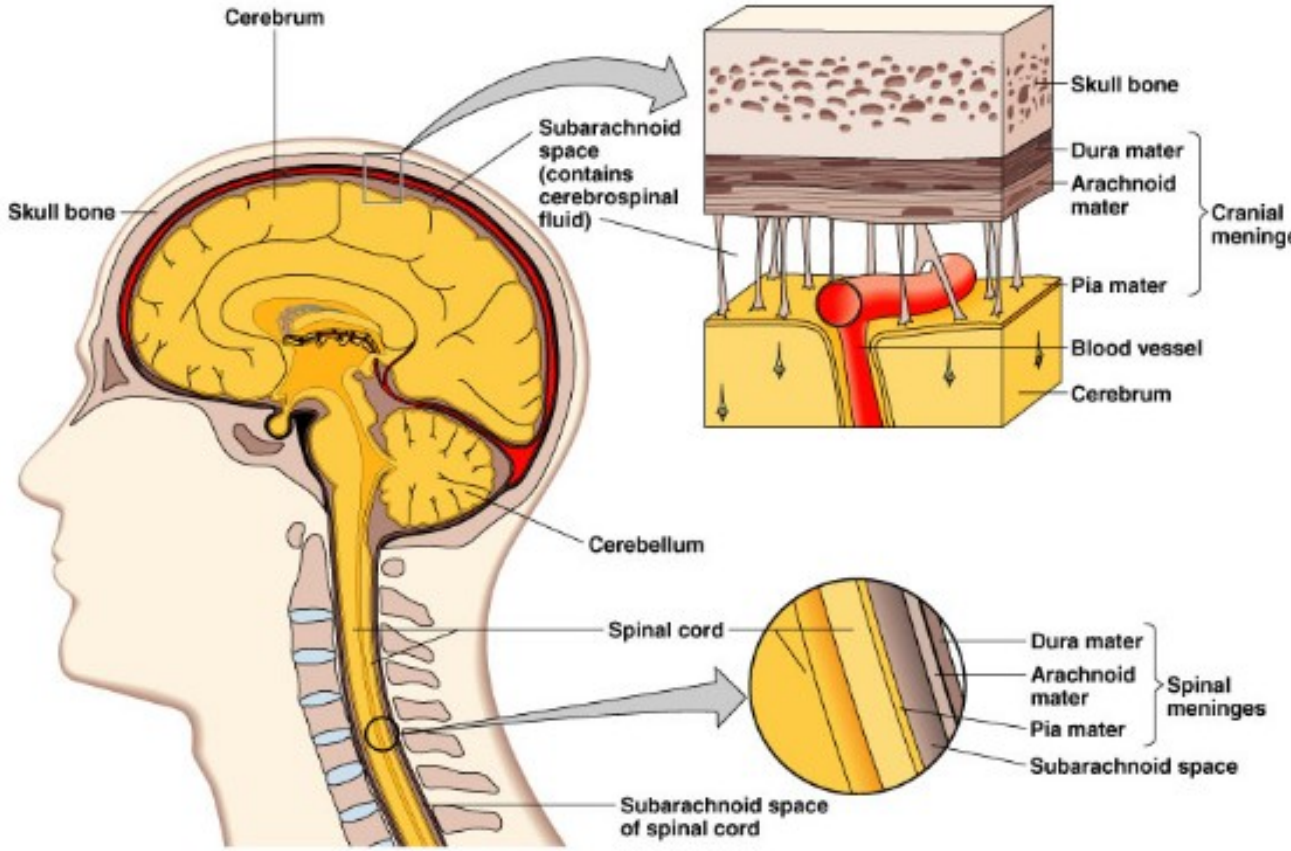
Central Nervous System

- Brain and Spinal Cord
- Interprets, Coordinates and Integrates
  - input from sensory nervous system
  - output to motor nervous system
- Delicate and Vital
- Fragile, does not heal or regenerate well
- 3 structures for protection:
  - *Bone*
  - *Meninges*
  - *Cerebro-Spinal Fluid (CSF)*

Learning Aim: Describe how the skull and vertebral column protect the nervous system.

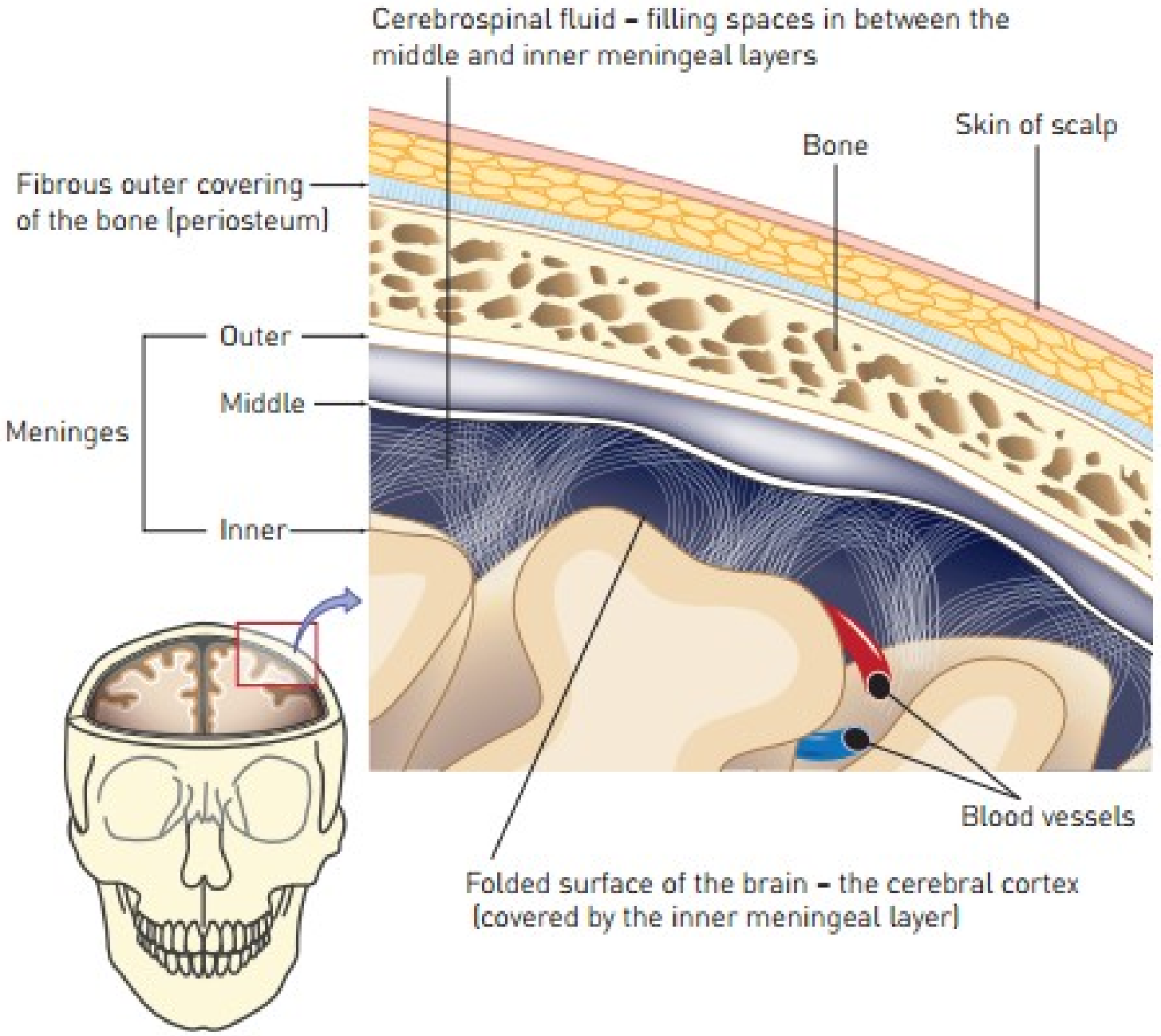
Protection of CNS

- **Bone**
  - Cranium (skull) houses the brain
  - Spinal cord within vertebral canal in vertebrae
  - Rigid structure to protect from penetrating and pressure injuries.



- **Meninges**
  - 3 layers of connective tissue
  - Between bone and CNS
  - Fibrous outer layer – dura mater "*tough mother*" – stuck to inside of skull.
  - Loose mesh middle layer – arachnoid mater – "*spider mother*" – allows for flow of CSF
  - Subarachnoid space – cerebrospinal fluid flows through this space.
  - Delicate inner layer – pia mater – "*gentle mother*"
    - Blood vessels
    - Sticks closely to surface of brain, and spinal cord

Learning Aim: Describe the structure of the meninges in the brain.



Learning Aim: Describe the structure of the meninges in the brain.

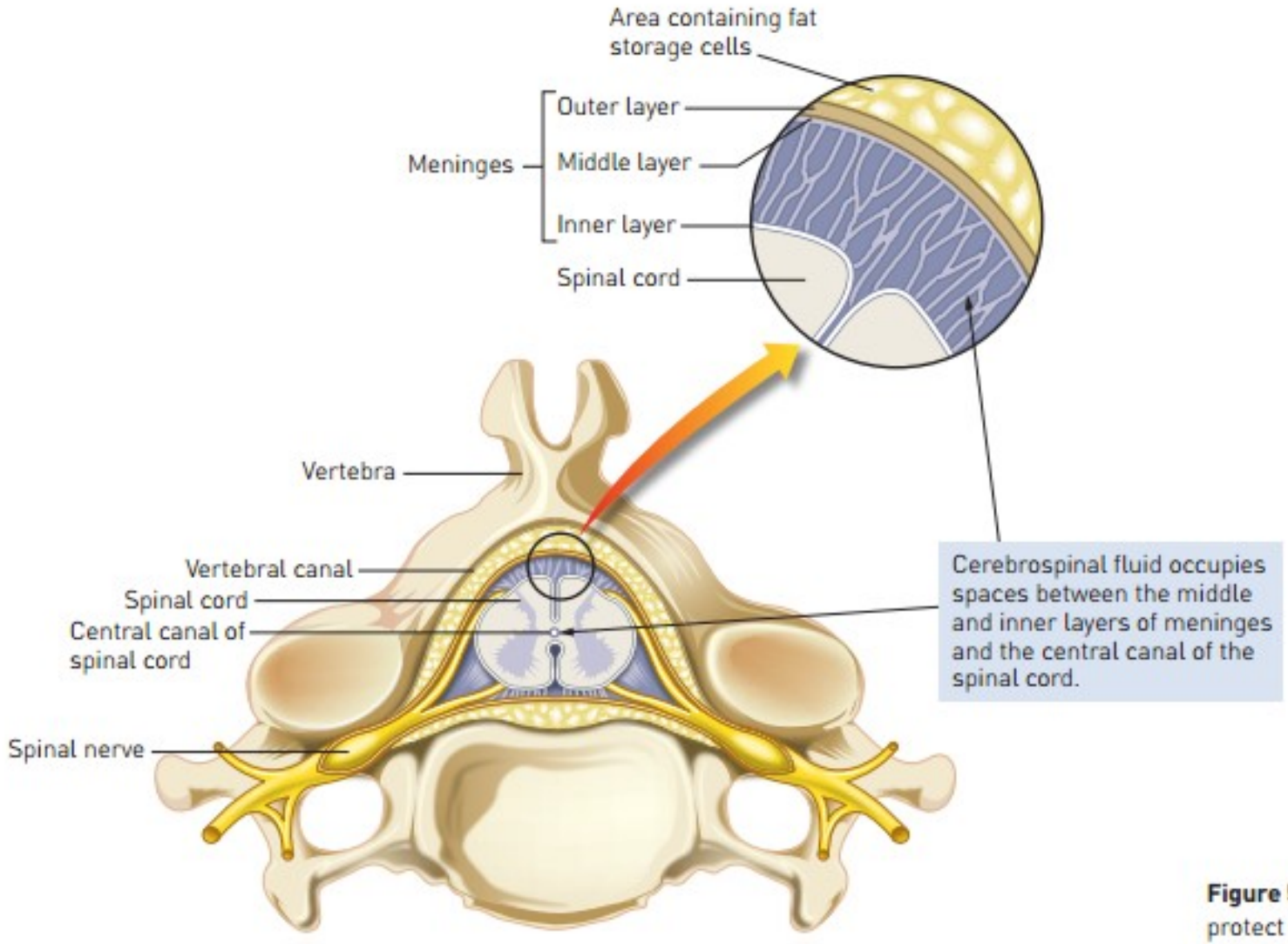
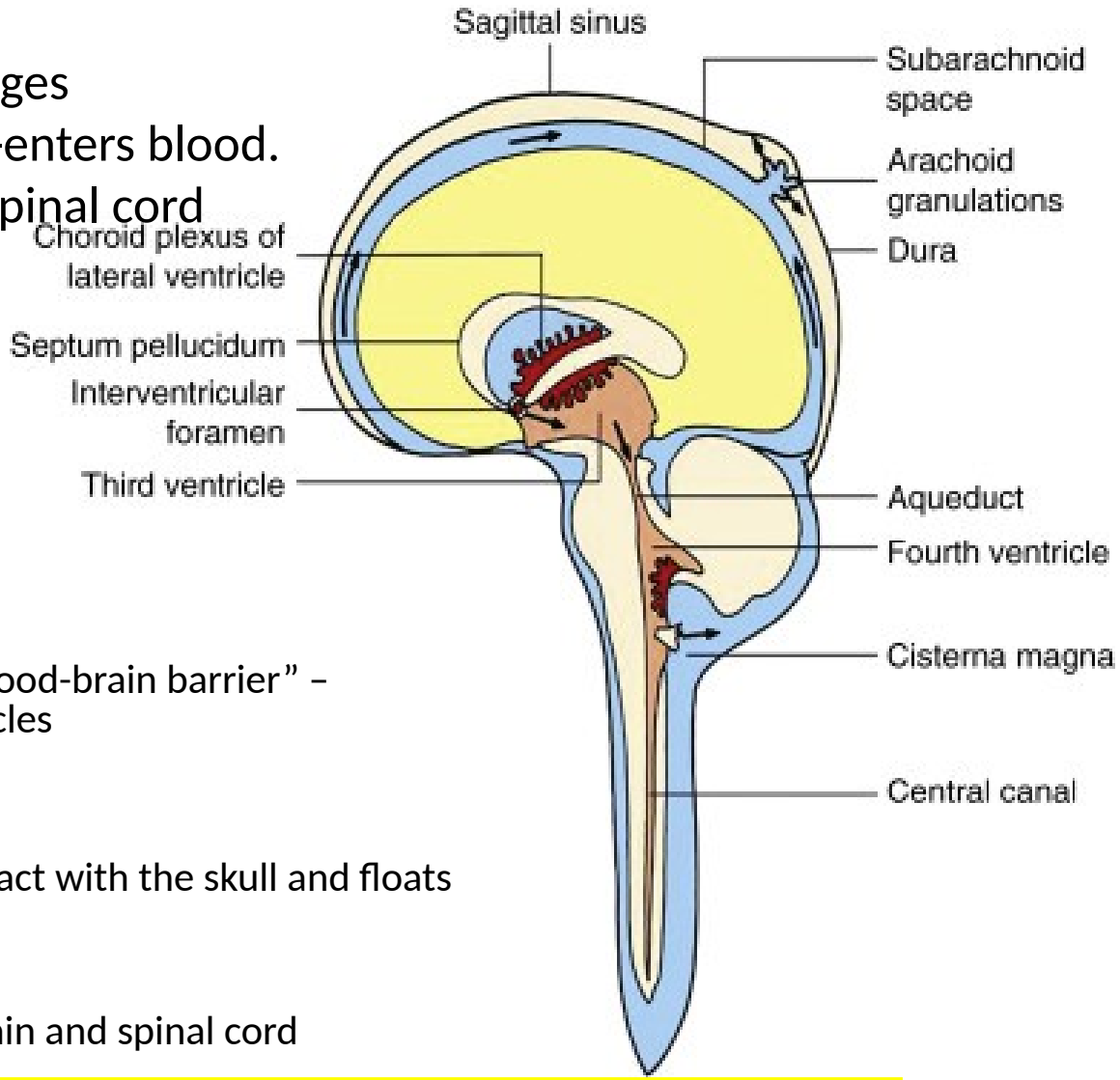


Figure 5.3 Structures that protect the spinal cord

Learning Aim: Describe the structure of the meninges in the spinal cord.

Protection of CNS

- **Cerebrospinal Fluid (CSF)**
  - Flows between middle and inner layers of meninges
  - Formed from blood, circulates and eventually re-enters blood.
  - Circulates through cavities in brain and canal in spinal cord
  - Clear, watery fluid:
    - Few cells
    - Glucose, protein, urea, salts



- **3 functions:**

Protection

- Shock absorber to protect from impact
- Barrier between blood vessels and CSF is quite tight “blood-brain barrier” – protects against entry of bacteria and some other particles

Support

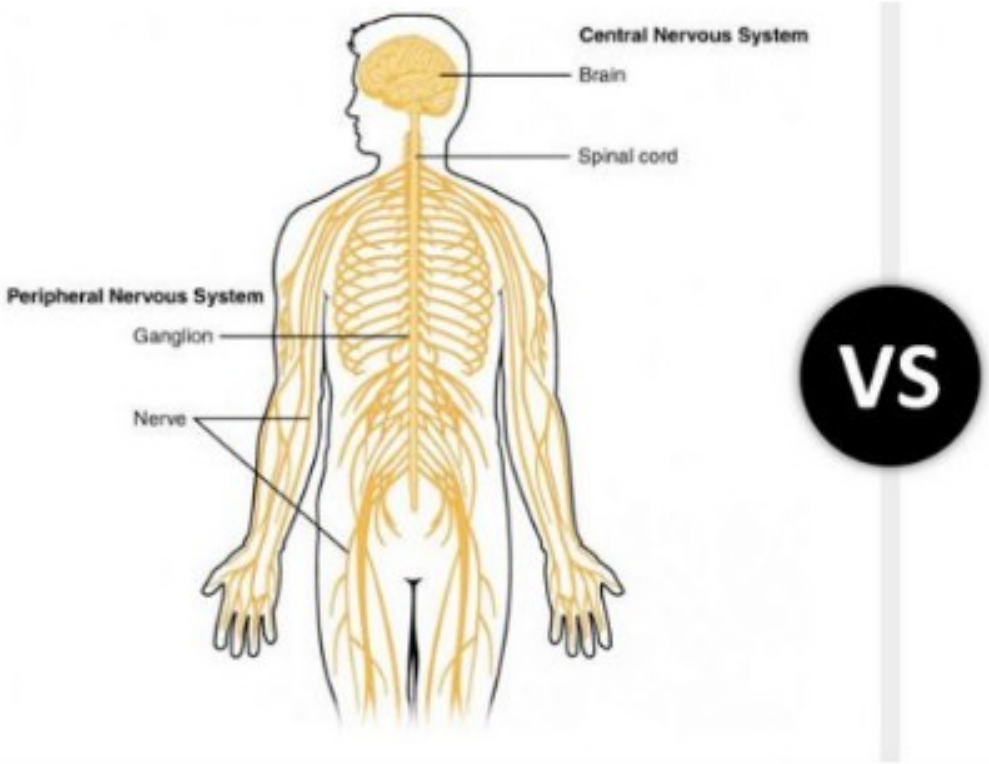
- Supports the fragile brain tissue – cushions it from contact with the skull and floats it within the cranial cavity

Transport

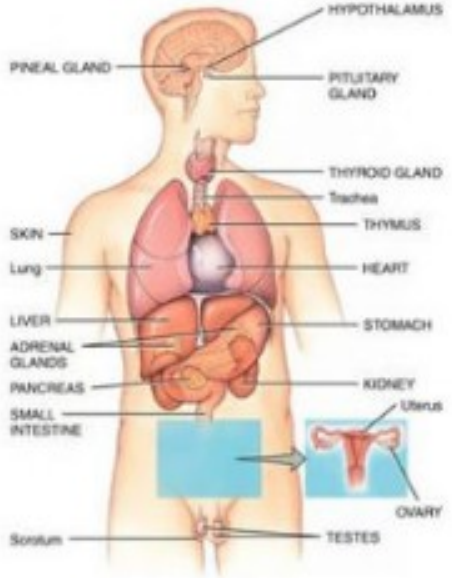
- Transports nutrients and wastes to and from cells of brain and spinal cord

Learning Aim: Describe the production and composition of the CNS, and discuss how the meninges protect the CNS.

Nervous and Endocrine Systems: Comparison



VS



Nervous and Endocrine System Comparison - Similarities

- Both involved in sending messages from one part of the body to another
- Some substances function as both neurotransmitters AND hormones eg:
  - Dopamine
  - ADH (Anti-Diuretic Hormone)
- The two systems interact, eg:
  - Hypothalamus receives neural messages and can respond by triggering an endocrine message.
  - Sympathetic NS simulation during fight or flight triggers release of adrenalin from adrenal medulla.
- Some hormones are secreted by neurons eg:
  - Oxytocin from Posterior Pituitary neurons
- Some hormones and neurotransmitters have similar effects on target cells/tissues
  - Noradrenalin and Glucagon both act on liver to cause glycogen to glucose.
  - Noradrenalin and Adrenaline have very similar effects on tissues.

Learning Aim: List similarities between the endocrine and nervous systems.

Nervous and Endocrine System Comparison - Differences

Table 4.3 A comparison of the nervous and endocrine systems

Characteristic	Nervous system	Endocrine system
Nature of message	Electrical impulses and neurotransmitters	Hormones
Transport of message	Along the membrane of neurons	By the bloodstream
Cells affected	Muscle and gland cells; other neurons	All body cells
Type of response	Usually local and specific	May be very general and widespread
Time taken to respond	Rapid – within milliseconds	Slower – from seconds to days
Duration of response	Brief – stops quickly when the stimulus stops	Longer lasting – may continue long after the stimulus has stopped

Learning Aim: List differences between the endocrine and nervous systems.